Tech Heroes #5321

“Goal-based groups, not role-based groups.”
“Labels on cables.” “#NotJustRobots”
“Striving for Activism and Inclusion”

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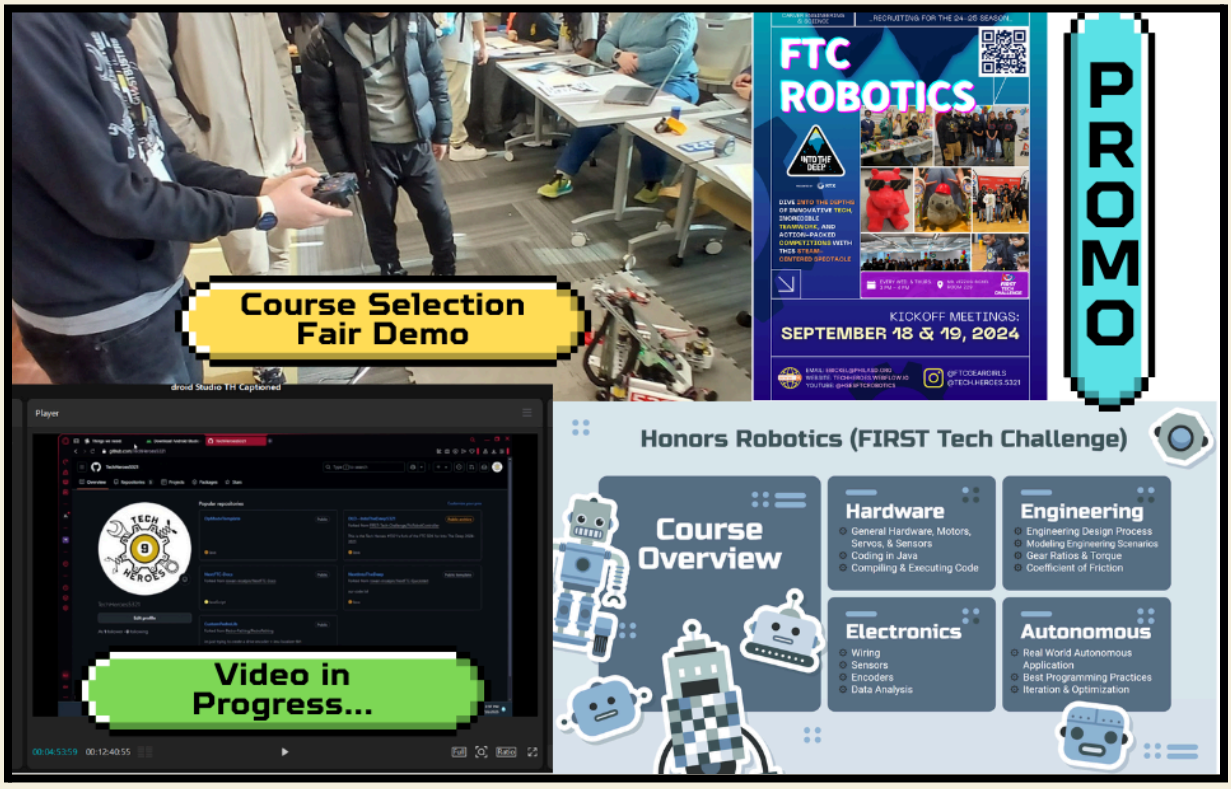
Team Plan

Season Goal:

- We are using this season of FTC to **prepare club members to form an FRC team in the fall**

FRC Preparation Goals:

- New practices implemented include:
  - **Switching from block coding to Java coding** in Android Studio
  - **Establishing a Business Team** for an additional funding backbone
  - **Implementation of CAD** through **learning FRC Design**
- **Purchasing better parts from GoBilda** to experiment with on this season's robot
- We **started a class for next year** giving FTC and FRC more time to work
  - We **introduced the basics of our class at our school's course selection fair** *(see left images)*.
  - **Students got to test drive and experiment with tuning the robot** for AUTO navigation *(see upper right image)*
- We posted an Android Studio setup tutorial on our YouTube channel for other teams & future members to see and express their thoughts on our methodology



AWARD COLOR CODING LEGEND

Innovate | Control | Motivate

Design | Think | Connect

Organizational Plan

Our Goals

The Tech Heroes aim to **create an excellent & inclusive environment for both returning members & newcomers alike.**

- Frequent discussion of plans and through both in-person & online communication (i.e. Discord servers, Instagram DMs, scheduled calls)
- Establishing **goal-based** groups
- Performing **2 week check-in cycles** to evaluate the team's progress as a whole through members sharing out from the goal-based groups they opt to work in

We want to open the FIRST community in our school

- We promoted **#NotJustRobots** in our recruitment efforts to spread awareness of our club in a non-STEM focused light.
- We also let **Middle Schoolers** join for the first time to introduce FIRST into a new community within our school.
- Next year we will be starting an **FRC Team** at our school and this year of FTC is to help us prepare

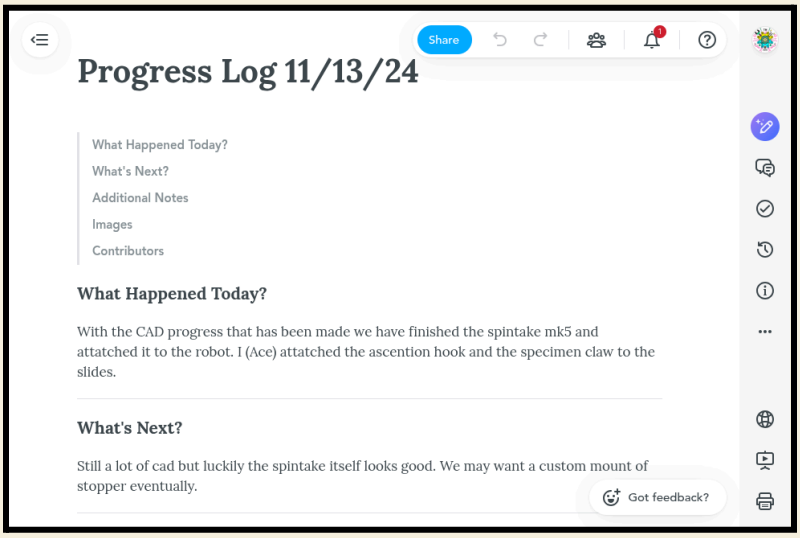
Finance Plan + Business Team Launch:

- Most of our funding comes from our local and school communities, particularly through successful bake sale fundraisers that allow us to buy new parts.
  - Our October bake sale netted us **$800+** allowing us to purchase new GoBilda parts that we used for this season's robot
- This year, we **joined forces with our sister team, Gear Girls, to form a business team to alleviate the loss in funding due to the new lack of funds from the school district**.
- We subsequently **reached out to HAAS Automation and our school's Home and School Association to earn roughly $2500 in grant funds**

Risk Management:

- We used CAD to create parts to prepare for any possible mechanical risks such as **3D printed spacers to circumvent inconvenient mounting** and **casings to protect the wheels.**
  - Creating these custom parts would've been impossible with our existing COTS parts
  - These custom parts **allow us to integrate COTS parts from different companies**
- Members will always coordinate with each other to avoid integration issues

Our Communications:



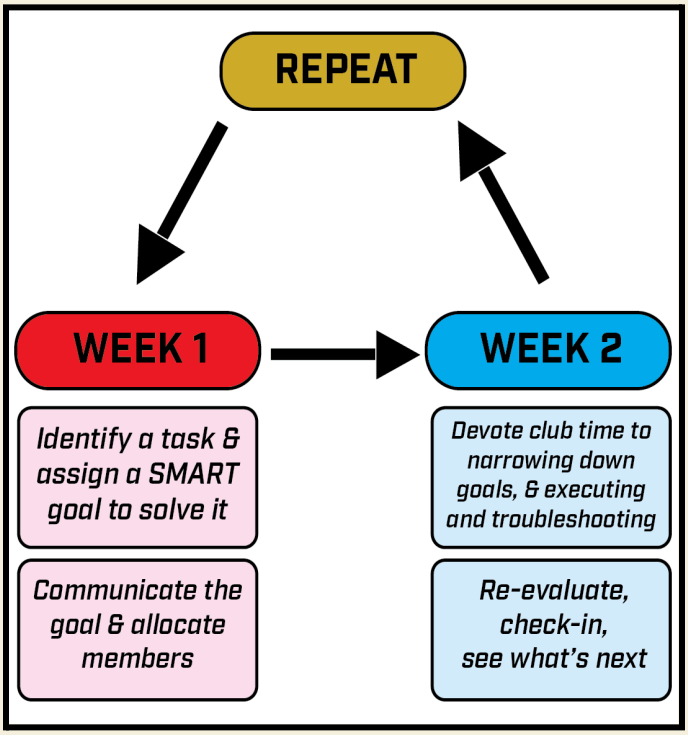
- **Google Classroom:** General announcements & Weekly Slides with 2 week cycle system
- **Discord:** Discussions and forums for subtopics like fundraising & future plans. Virtual calls to discuss deadlines and huge changes.
- **Instagram:** Share video & photo highlights from our season; connect with teams via DMs to discuss robot strategies and engineering portfolios
- **ZOOM:** Coordinate virtual calls with global teams to discuss strategies, outreach, etc
- **MeisterNote:** Used for taking and organizing logs, created note-taking templates for new recruits to adopt

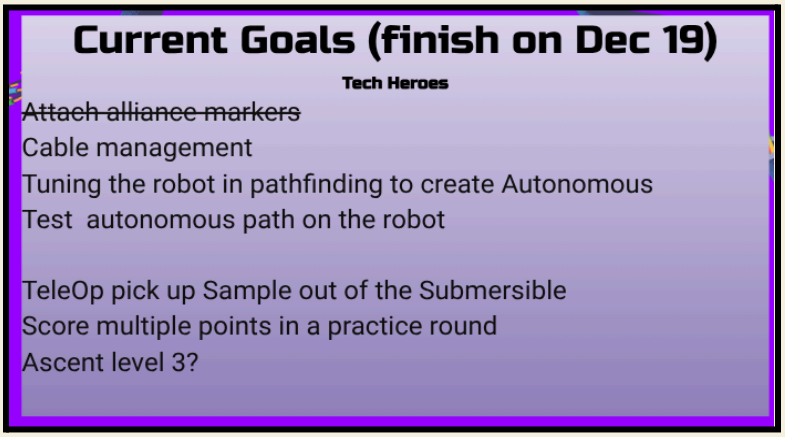
Think: Engineering & Design Process (Goal-Based Groups)

Two Week Cycle + SMART Goals:

SPECIFIC	MEASURABLE	ACHIEVABLE	RELEVANT	TIME-BASED
Target particular problems for improvement	*Ensure processes involved in goals are trackable*	*Ensure goals are not too grandiose*	*Ensure goals serve a purpose that connects back to the relevance of our team and principles*	*Ensure goals can be completed by a dedicated deadline*

- **SMART Goal-based groups** are established, updated, & actively pursued based on the tasks we decide upon in our **2 week cycle system**
  - **Based on our initial robot design, our primary groups were given a large overall task that had to be completed by the time we compete** ***(see page 6)***
- 2 week cycles divide the larger goals into more defined tasks that are checked up on & updated continuously
  - See diagram on right and example below for more details





Connect: Gracious Professionalism

- We share **Provided Published Resources** on our **website and YouTube Channel** such as **informative documentation, 3D models, and competition resources**.
  - Although the site is intended for FIRST teams, it is **publicly available**, which means **anyone who is remotely interested in robotics and/or our team philosophy can readily access it**.
  - Our **shared YouTube Channel with Gear Girls features tutorials of the fundamentals of mechanical components based on our understanding and testing from years prior**.

Motivate/Connect: Recruitment Efforts & STEM Promotion

Summer Bridge:

- During the summer (August 2024), our school holds a week long summer program for **incoming underclassmen**
- **We presented to them** on the second day *[see page 7 for more details]*

Town Halls:

- Our school holds monthly town hall meetings to fill the students in on incoming events
- **We present to the freshmen and middle school students** to advertise ourselves

Back to School Bash:

- During our annual club fair, we ran a table & got **60+ sign ups**
- We had a **slideshow displaying our accomplishments** running
- Last year's robot, **Shifty, was driving with a sign on him**
- **We engaged with students to answer questions**

Kickoff Stations:

- We came up with a series of stations to throw new members directly into what robotics is all about
- Our stations were:
  - **Building**
  - **Coding**
  - **Sketching/CAD/Design**
  - **Game Demo/Driving/Strategizing**
  - **Q&A**
- People were able to easily dive in to whatever part of the club they were interested in which encouraged them to join despite prior tech knowledge

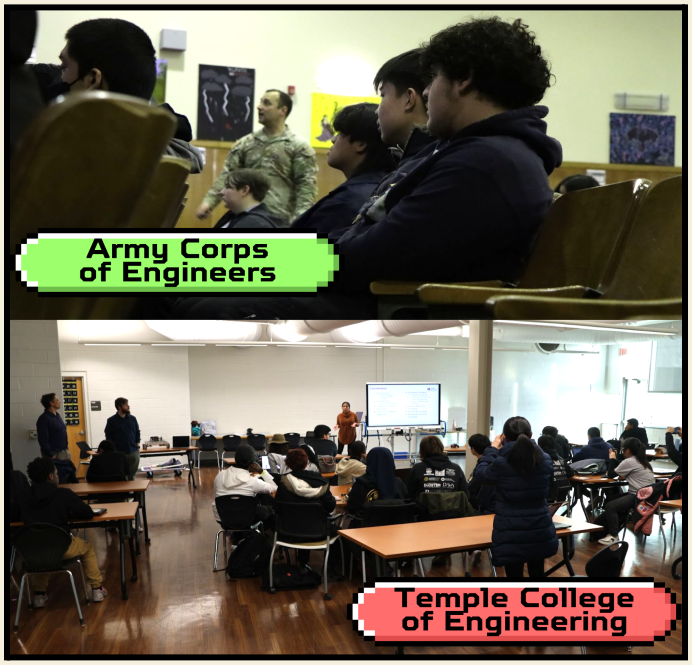
- They could also rotate stations to try new things, allowing members to diversify and gain new skills

Neubauer Family Foundation:

- Our school is planning on implementing a dedicated STEM makerspace in some unused areas in our school
  - Members of our schools community testified to the eye-opening impact STEM has on us
  - Groups present included veteran Tech Heroes Members, the founder of the National Society of Black Engineers Jr Chapter (NSBE Jr.) at our school, and our school administration
  - The Foundation was impressed by our experiences in FIRST

Outreach Events

- **Temple Engineering Visit**
  - Developed strong connections with staff
  - Giving members connections for **college and career exploration**
  - **Planned tour of Temple's College of Engineering** in the near future to augment the connection
- **Army Corp. of Engineers**
  - Some representatives **presented their environmental projects and skill management** & questionnaire for continued interest
  - **Took representatives on a tour** around our school to visit our computer science and chemistry classrooms
  - **Planned tour of their facilities** in the near future to augment the connection



- **Nautilus Virtual Visit**
  - Presentation to learn about their aquatic work in the STEM and robotics space
  - Concluded with questionnaire session for continued interest in their program and practices

Meeting with Local & International Teams:

- We got into contact with an **England-based rookie FTC team, Cassiopeia UK625.** We **coordinated a virtual call and gave them tips on their mechanics, engineering portfolio, team management, etc.**
- We also **met up with fellow FTC teams** from **Lincoln High School** and **Central High School** before Qualifiers to scout, talk strategy, and practice.

Mentors

- This year, we acquired a mentor, Mason.
- His **experience in the tech analytical space was instrumental in helping us in any way he could**: from navigating mechanical issues and coding problems, to finding sponsorships and applying for grants.

Motivate/Connect: Our Activism & Inclusion Efforts

Executive Student Action Committee:

- A few of our members opted to join the Philadelphia Robotics Coalition's Executive Action Committee (ESAC). ESAC focused on the following:
  - **Giving advice and feedback** to the Philadelphia Robotics Coalition (PRC) and its board members
  - **Leading Affinity Groups targeted toward communities that are underrepresented in STEM fields**
  - **Community Outreach through assisting FLL teams**, and **discussing STEM at other after school extracurricular events, like chess tournaments**

FLL Volunteering:

- Mentored several teams at an FLL qualifier and lead a rookie team to get an award
- Prepared and watched FLL teams during judging
- Ran a table at a youth chess tournament/STEM fair to promote the PRC

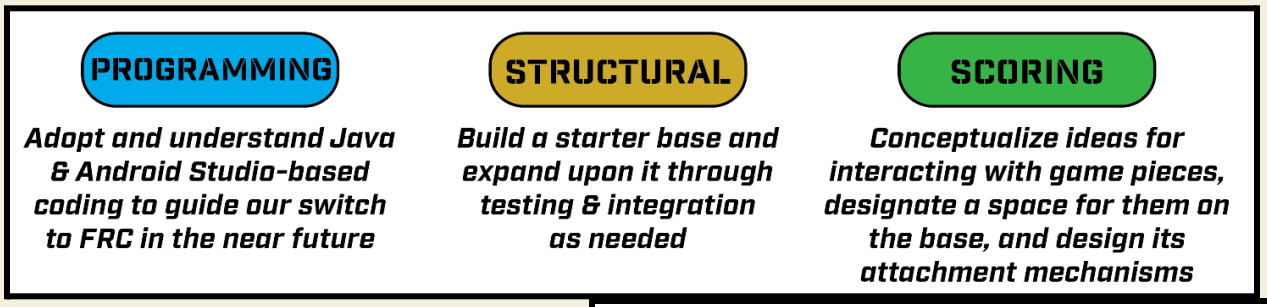
Advocacy at a School District Board Meeting:

- On October 24th, some of our members joined robotics teams and clubs across the city in peacefully protesting at the School District of Philadelphia Board Meeting that day. **We spoke about how their recent decisions regarding funding cuts for extracurriculars would limit the youth's access to STEM education.**
- We were successful in convincing the school board that they made a poor choice and now **more extracurricular funding is being reinstated**

Think: Our Goal-Based Journey

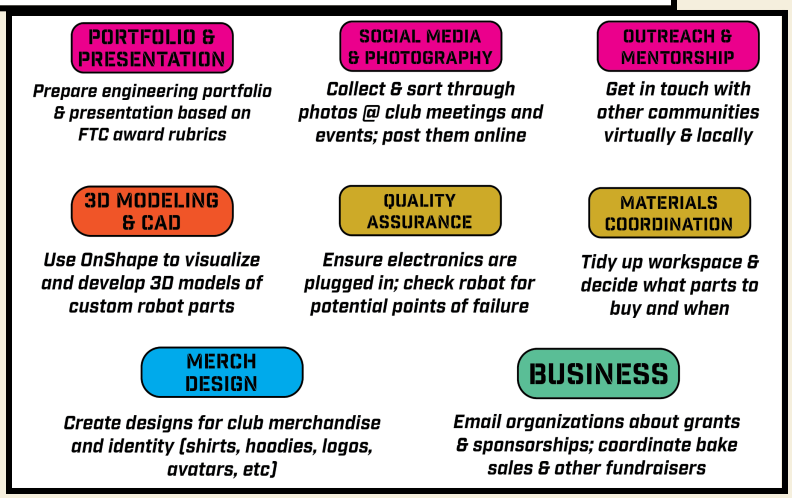
Transitioning to Primary Groups:

- Our kickoff stations evolved into the following primary groups to avoid integration issues.



Establishing Alternate Groups:

- Focus is to **build and maintain the structure and presence of our team as a whole beyond the robot itself**
- See diagram on the right for more details about specific groups and their tasks.



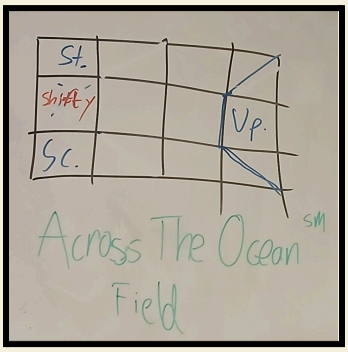
Defined Timeline:

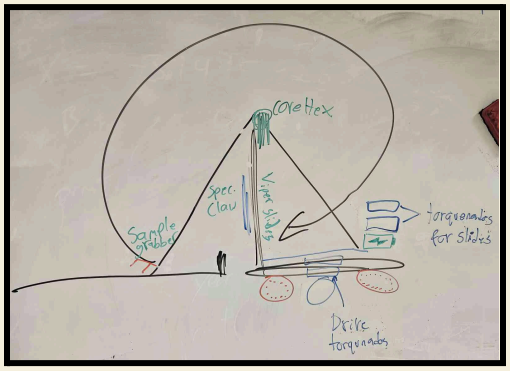
Below is a timeline of our progress for this season leading up to today.

Prelude:
Recruitment Rally

- **Introduced FTC to prospective underclassmen** at school's Summer Bridge Program & Town Hall presentations
- Planned **interactive, station-based kickoff meeting(s)** using previous season's robot *(left image)*
- Drafted initial ideas for robot navigation and design *(right image)*

Visuals:





MOTIVATE

Phase 1:
The True Kickoff!

- Back To School Bash!
  - **70+ interested students from grades 7-12 in attendance at kickoff!**
  - Students tinkered with the FTC game elements
- Alumni from Temple Engineering visited
  - **Showcased 3D printed projects to our principal; inspired new members to explore design**

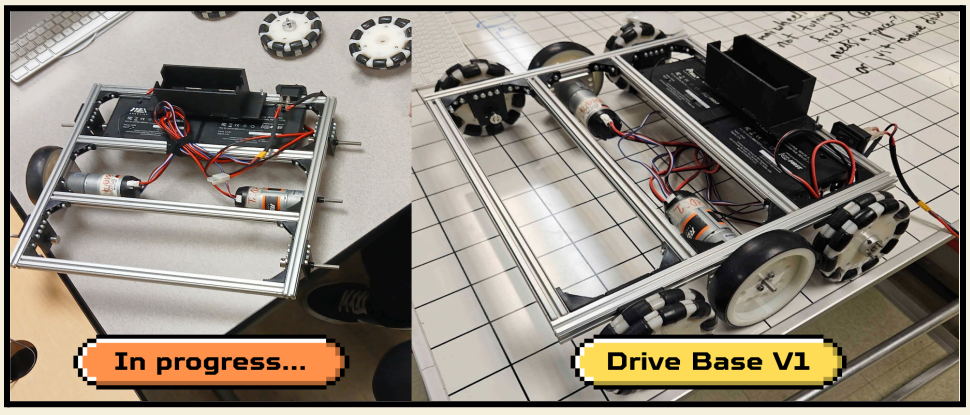
Visuals:

CONCEPT

Phase 2:
Touching Base

- **Defined robot goals with *Gear Girls* (sister team)**
- Opted for **fast, lightweight design with 6 wheels** (2 powered, 4 omniwheels); easy tank steering design
- Decided to redesign due to issues

Visuals:

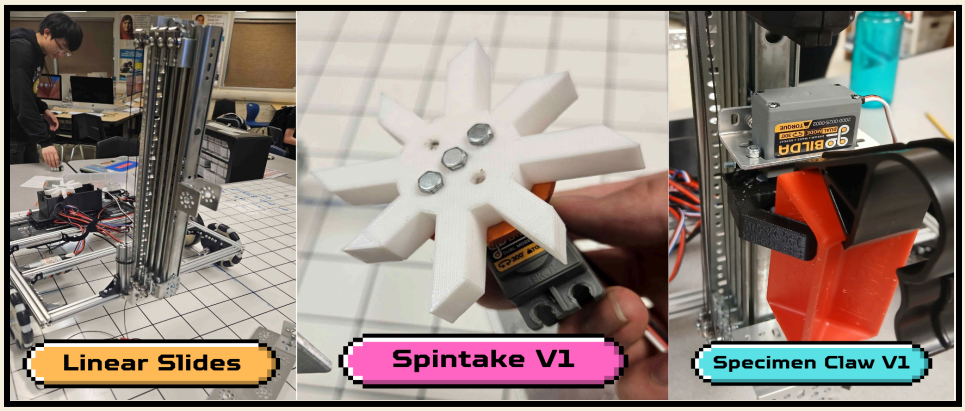


DESIGN

Phase 3:
Manipulator Prototypes

- Prototyped manipulator with **GoBilda linear slides** and **3D printed specimen claw & spintake (spinning intake)**
  - **Spintake funnels in samples to be scored in baskets**
  - **Claw attaches & detaches specimens from submersible**

Visuals:



INNOVATE

Interlude:
Finance Week!

- **Cornell alumni sponsored GoBilda parts** for our team's robot
- **Advocated for increased funding at district board meeting** ***(see page 6)***
- Wrote grants for various sponsors (HSA & HAAS CNC); **earned $2500!!**
  - **HSA grant publicized team ideals via Instagram video!**

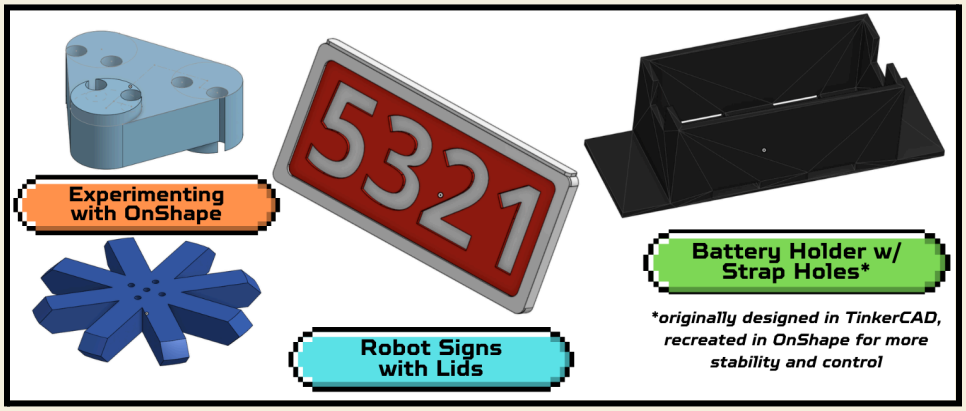
Visuals:

CONNECT

Phase 4:
CAD Week!

- We made lots of progress in CAD using **FRC Design**
- Designed **custom robot signs & holders** that are **easily swappable**
- Designed **custom stabilizer for sample pick-up assistance**

Visuals:



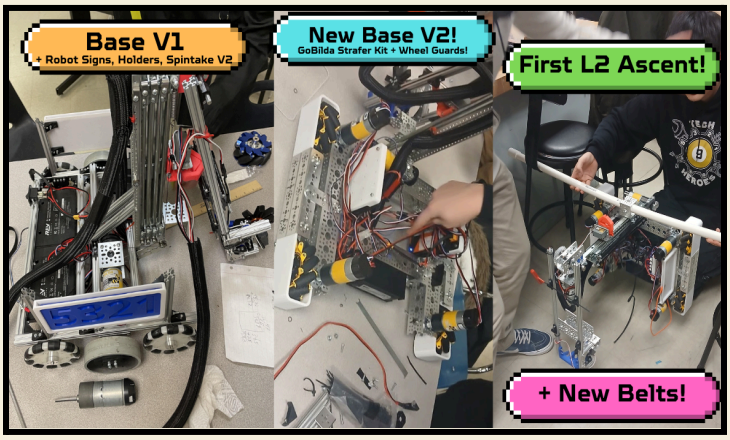
DESIGN

THINK

Phase 5:
Strafing into Success

- **Switched base to GripForce mecanum wheels** for better traction & smoother rotation
  - Reinforced wheels with **custom 3D printed casings for protection**
- **Revised spintake & claw designs** for more consistent sample/specimen manipulation
- **Replaced linear slide strings with belts** for **improved reliability and stability**
- Discovered **insufficient torque in arm motor, prompting a switch to a stronger motor using a complex bevel gear system**
  - Achieved Level 2 ascension!

Visuals:



CONNECT

Phase 6:
Elevations in Outreach

- Held a team call to discuss our deadlines
- **Contacted a team from London with Gear Girls**
- Advertised our upcoming FTC class at school Course Selection Fair
- Attended an event held by Central HS & Lincoln HS to test our robot on an actual field, and discuss robot strategies
- **Reconnected with local women-run apparel organization Boxed Sourcing for club t-shirts**

Visuals:

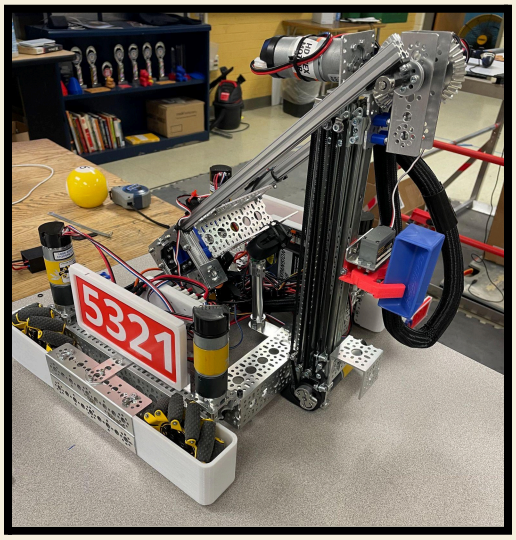
DESIGN

Phase 7:
The Final Stretch!

- Built the finalized motor mount for the hypotenuse arm
- Streamlined code for easier navigation and proofreading
- Re-achieved Level 2 ascent using hypotenuse crossbar!
- Added plushie for hypotenuse arm cushioning
- Posted Android Studio setup tutorial on YouTube channel!

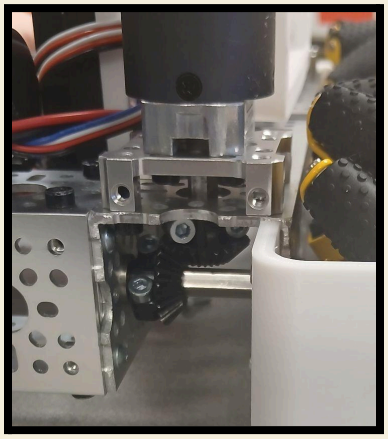
Visuals:

Design: Euler: Herald of the Hypotenuse

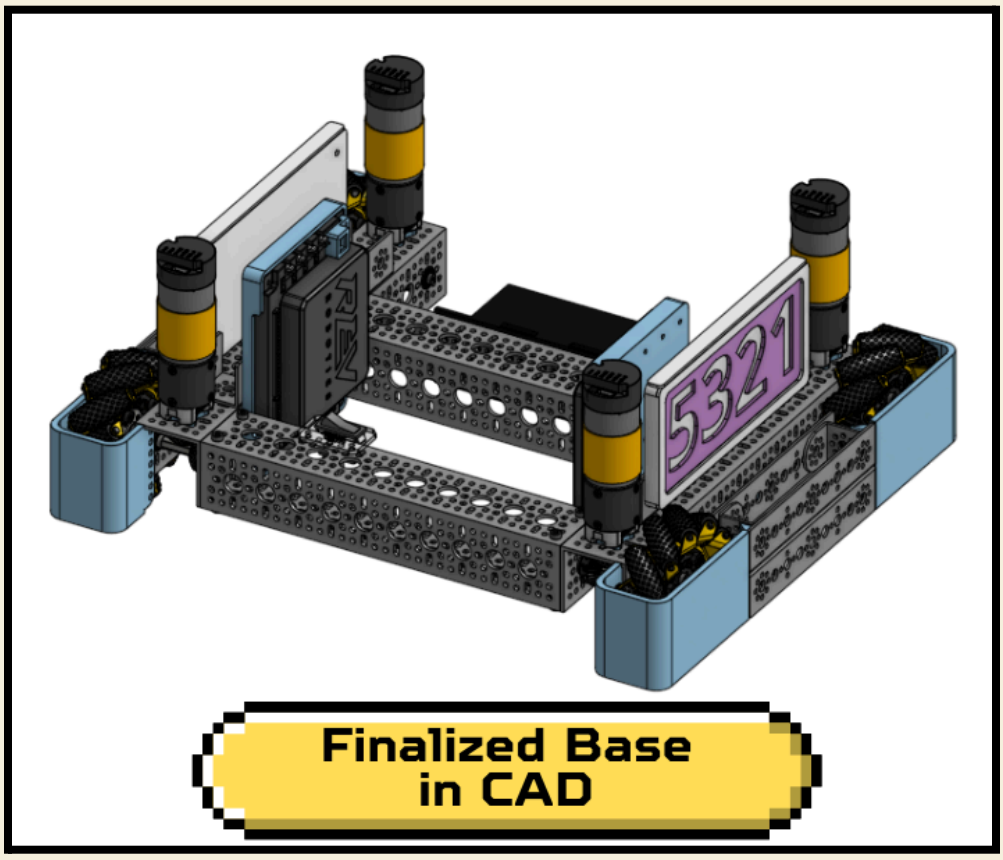


- This is Euler (pronounced oil-urr)
- Name originated from the **angled arm (hypotenuse arm) on the front of the robot**
- We love the innovation (and mathy references) we put into it, including its field-oriented driving and custom-made 3D printed appendages!

Design/Think: Drive Train



- Euler uses a GoBilda **GripForce strafer kit for better traction & smoother rotation** as compared to our former 6 wheel west coast drive design
  - We have **3D printed wheel casings for protection and sleekness,** they also allow us to safely slam into things
  - **Beveled up motors for space conservation** *(see image on right)*
  - New GoBilda yellow-jacket motors that have 312 RPM
- We **made the entire base in CAD first to troubleshoot any issues** before we building it
  - We learned this was a **much more time-efficient way of designing**

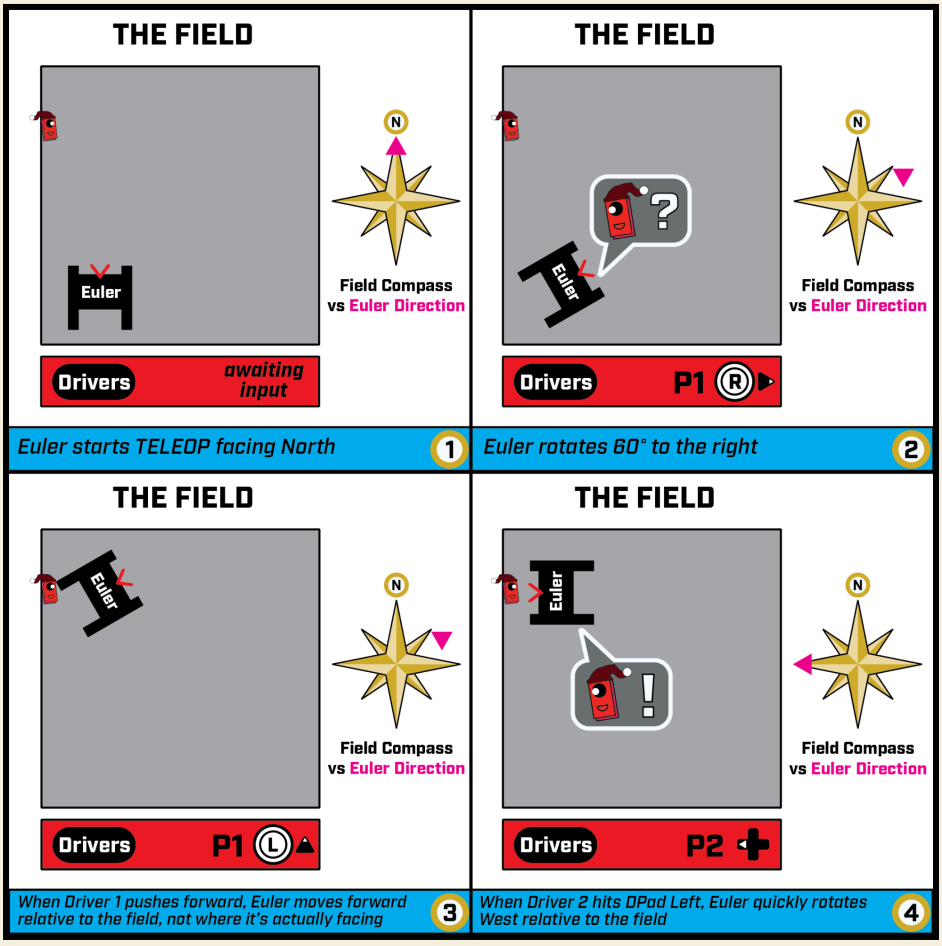


Control: Adopting Java & Android Studio

- We learned how to navigate **GitHub** repositories and use **Android Studio** to do more advanced programming
- As part of our switch **we worked with programmers at Google** to create a webpage of the blocks editor
- We used a series of libraries to code similarly to FRC and improve the performance of our robot:
  - **Pedro Pathing** - A path following library
  - **Dashboard** - A driver station on our computer for editing variables live
  - **NextFTC** - A beta command based library
- For **Pedro Pathing**, we created a new localizer (software that tells the robot where it is and where it's facing) using the drive encoders and the Inertial Measurement Unit (IMU) to be more accessible to most teams as opposed to more expensive solutions like odometry
- We are a very early adopter of **NextFTC**, in fact we are one of their first java users
- We coded the hypotenuse to resist gravity equally at all points using physics and trigonometry
- We used several complex encoded motor controller algorithms for smoother, more efficient use:
  - **Proportional Integral Derivative (PID)** - Moves and holds motor in place, preventing overshoot
  - **Gravity FeedForward** - Powers the slide motor to resist gravity
  - **Arm FeedForward** - Resists gravity according to arm position
- In order to help many people switch like us, we made a YouTube tutorial on installing Android Studio

Field-Oriented Driving

- We **changed to a field-oriented drive scheme** when we switched to the GripForce base
- Field-oriented driving relies on the IMU and remains accurate no matter what
- **More intuitive** as Euler moves forward **relative to driver's POV of the field**
- We use "compass directions" to help clarify Euler's rotation
- Player 2 can use the Dpad to set Euler to rotate to a compass direction



Innovate/Design: Manipulators in Rotation

Spintake

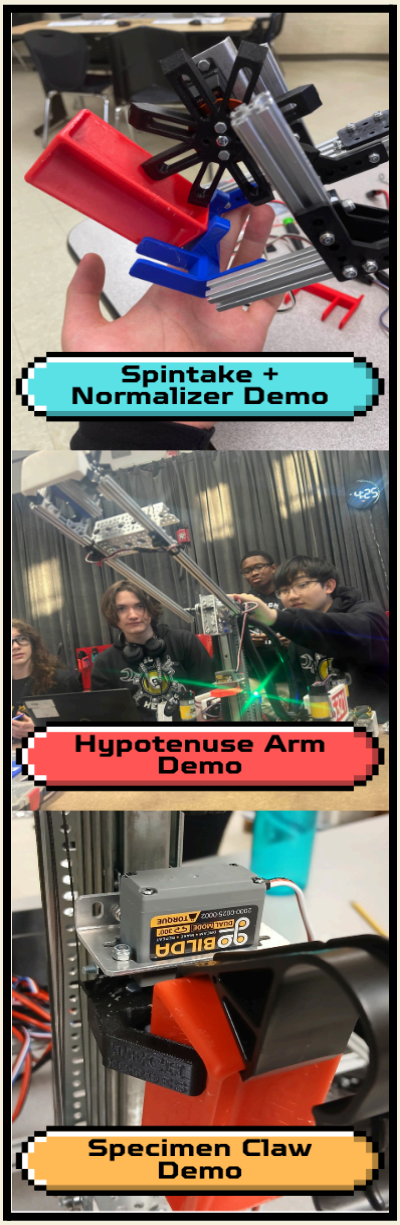
- Spinning wheel against a stabilizing wall called **The Normalizer** *(see page 14)*
  - Funnels samples in & pinches in place
- Doesn't require the precision of a claw for manipulating samples
- Easily removable and replaceable
- Rests at the end of **hypotenuse arm** *(see below)*

Hypotenuse Arm

- Rotating arm
  - Reaches low to get samples off the ground
  - Drops high to reach the high basket
- Rests at an angle
  - Reaches forward in the submersible for effective sample grabbing
- Multipurpose arm will conserve space
  - Integrated crossbar allows for safer Level 2 ascent *(see page 9, phase 7)*

Specimen Claw

- Custom made pinching mechanism
  - Used for grabbing specimens from field walls
- Aided by linear slide mechanism for scoring specimens on high rung



Design: Conquering CAD

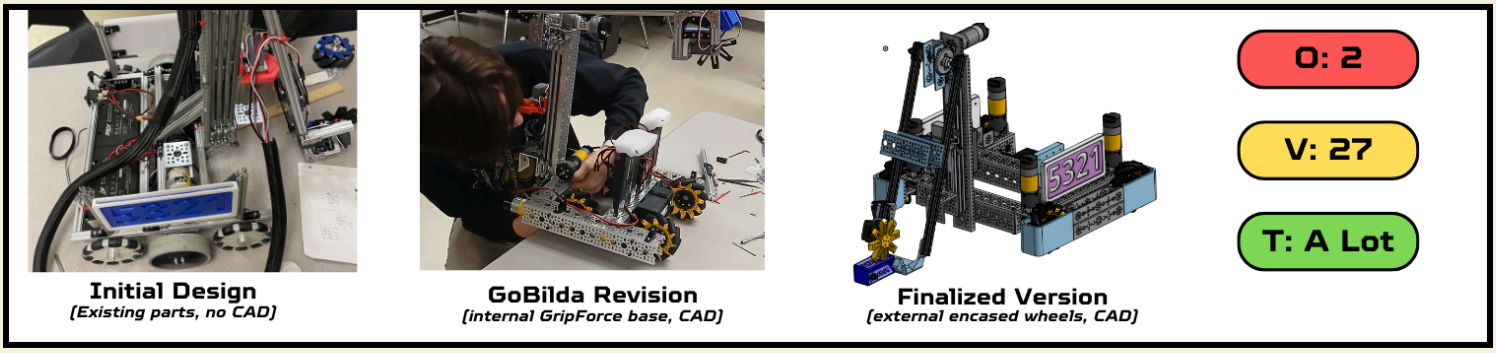
Statistics:

- Our final robot has:
  - **51 - 3D printed parts**
  - **6 - modified parts**

Vocabulary:

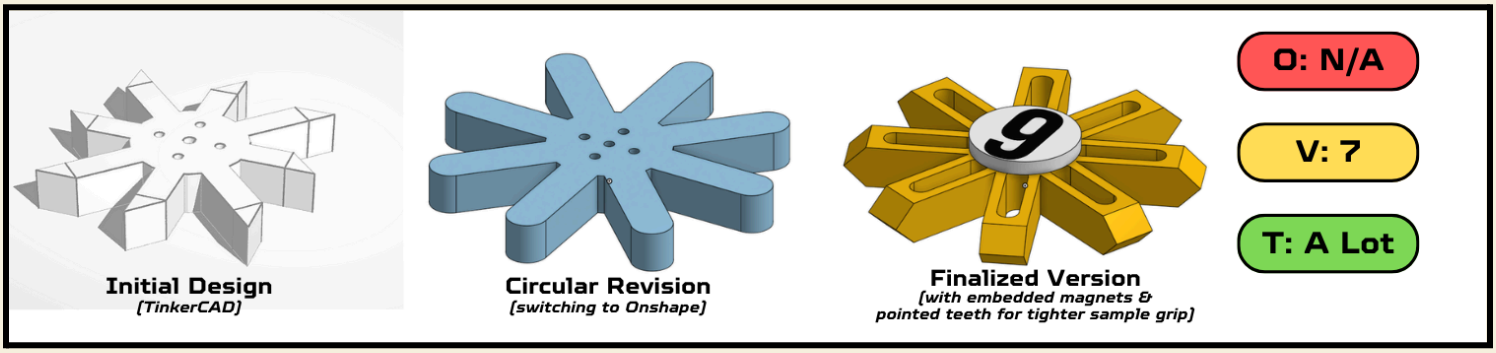
- To define degrees of change in our 3D printed robot addendums, we established the following vocabulary called OVT (Overhaul-Version-Tweak):
  - **Overhauls:** Complete redesigns that fundamentally change robot functionality and require reprints or rebuilds
  - **Versions:** Somewhat major changes that often require real world testing and changes to existing assemblies
  - **Tweaks:** Minor changes that might not require a reprint/rebuild, typically smaller part modifications and are not easily countable

Base Evolution:



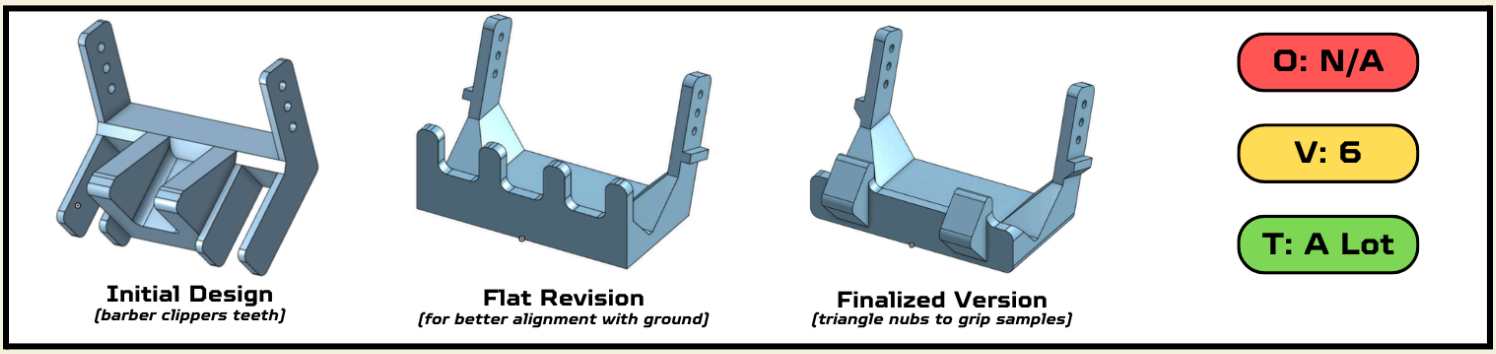
The base had **2 major overhauls** *(shown above)* moving from REV to internal GoBilda to external GoBilda with 3D printed wheel guards.

Spintake Evolution:



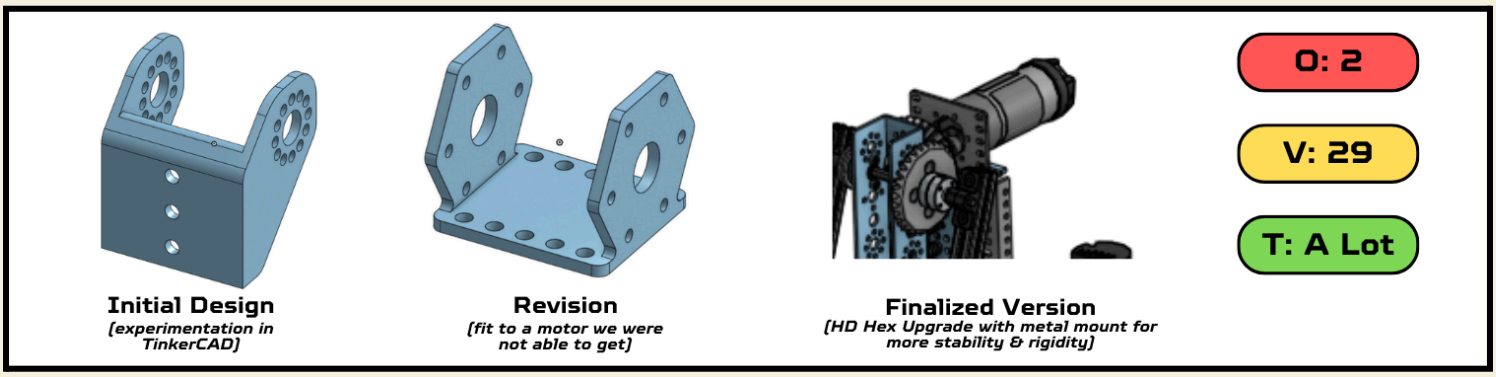
The spintake **did not have any overhauls**, as it fundamentally stayed the same from start to finish. The spintake went through **7 versions** and **many tweaks** *(shown above)*.

Normalizer Evolution:



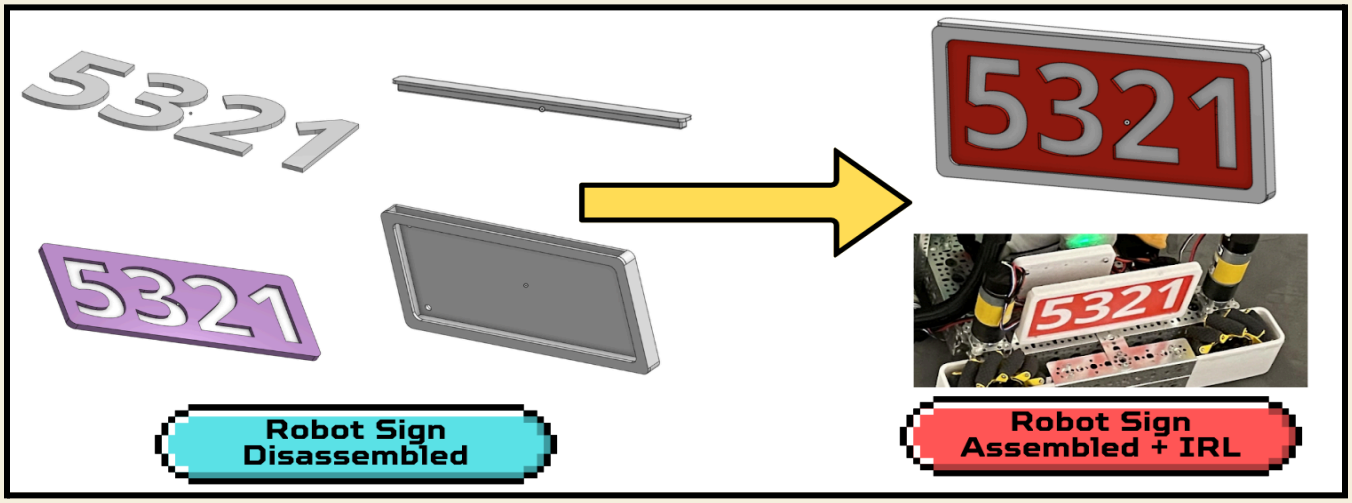
The Normalizer went through a lot of changes (including **6 versions**), with many of them breaking along the way. This ended up being one of the most troubling parts we had due to the hypotenuse arm's sheer strength.

Hypotenuse Motor Holder Evolution:



Before we switched to an HD Hex Motor with metal reinforcements for our hypotenuse arm, we used a 3D printed motor holder to hold the less efficient Core Hex Motor in place. This part had **2 overhauls**. At one point we wanted to use a REV Ultraplanetary motor with 90° gearbox but we couldn't get those parts *(see middle image)*.

Custom Robot Signs & Holders:



We opted to not only 3D print our robot signs, but also create customized holders that would easily slot the signs in place. Additionally, the holders can be sealed to prevent the signs from falling out during matches.